

CONTACT INFORMATION	311 Ferst Drive, Room 1114 Atlanta, GA 30332-0340 https://yz3062.github.io/	323-633-2456 yzhou3003@gatech.edu yuxin.zhou@whoi.edu
EDUCATION	Columbia University , New York, NY Ph.D., Earth and Environmental Sciences Dissertation: Atlantic Meridional Overturning Circulation instabilities during the last glacial cycle. Advisor: Jerry F. McManus University of Southern California , Los Angeles, CA B.S., Geological Sciences, <i>Phi Beta Kappa</i> Minor in Computer Science	2016 – 2022 2012 – 2016
EXPERIENCE	Postdoctoral Fellow, Georgia Institute of Technology Advisor: Jean Lynch-Stieglitz Guest Investigator, Woods Hole Oceanographic Institution Advisor: Geoffrey Gebbie Postdoctoral Scholar, UC Santa Barbara Advisor: Lorraine Lisiecki	2025 – 2025 – 2022 – 2025
PEER-REVIEWED PUBLICATIONS	20. Creel, R.C., Kopp, R.E., Dutton, A., Zhou, Y., Raymo, M., Britt, C., and DeConto, R. (2026) North American ice sheet persistence into past interglacials should inform future projections. <i>Nature Communications</i>. In press. 19. Zhou, Y., Lisiecki, L.E., Meyers, S.R., Lee, T., and Lawrence, C.E. (2026) Global and regional Pleistocene benthic $\delta^{18}\text{O}$ stacks on age models with and without orbital tuning. <i>Geochronology</i>. 8, 85–107. doi:10.5194/gchron-8-85-2026 18. Rowland, G.H., Hendry, K.R., Annett, A.L., Ng, H.C., Robinson, L.F., Sherrell, R.M., Zhou, Y., McManus, J.F., Brearley, J.A., and Li, T. (2025). High lithogenic fluxes at the West Greenland margin traced by $^{232}\text{Th}/^{230}\text{Th}$ in seawater and sediments. <i>Global Biogeochemical Cycles</i>. 39, e2025GB008531. doi:10.1029/2025GB008531 17. Zhou, Y., McManus, J.F., Pallone, C., Kenna, T.C., Weinstein, G.A., and Garcia, H. (2025). Abrupt weakening of deep Atlantic circulation at the last glacial inception. <i>Nature Communications</i>. 16, 7555. doi:10.1038/s41467-025-62960-y	

16. Hoogakker, B.A.A., Davis, K., Wang, Y., [and 57 others, including **Zhou, Y.**] (2025). Review of proxies for low-oxygen paleoceanographic reconstructions. *Biogeosciences*, 22, 863–957. doi:[10.5194/bg-22-863-2025](https://doi.org/10.5194/bg-22-863-2025)
15. **Zhou, Y.**, Lisiecki, L.E., Lee, T., Gebbie, G., and Lawrence, C.E. (2024). Regional benthic $\delta^{18}\text{O}$ stacks for the “41-kyr world” - an Atlantic-Pacific divergence between 1.8-1.9 Ma. *Geophysical Research Letters*, 51, e2023GL107858. doi:[10.1029/2023GL107858](https://doi.org/10.1029/2023GL107858)
14. **Zhou, Y.** and McManus, J.F. (2024). Heinrich event ice discharge and the fate of the Atlantic Meridional Overturning Circulation. *Science*, 384(6699), 983-986. doi:[10.1126/science.adh8369](https://doi.org/10.1126/science.adh8369)
Media Coverage: [Eos](#), [Live Science](#), [National Geographic En Español](#), [La Presse](#)
13. Zeng, M., Rashid, H., **Zhou, Y.**, McManus, J.F., and Wang, Y. (2023). Dynamics of the subpolar gyre and transition zone of the North Atlantic during the last glacial cycle. *Quaternary Science Reviews*, 314, 108215. doi:[10.1016/j.quascirev.2023.108215](https://doi.org/10.1016/j.quascirev.2023.108215)
12. **Zhou, Y.** and McManus, J.F. (2023). Authigenic uranium deposition in the glacial North Atlantic - implications for oxygenation, carbon storage, and deep water mass geometry. *Quaternary Science Reviews*, 300, 107914. doi:[10.1016/j.quascirev.2022.107914](https://doi.org/10.1016/j.quascirev.2022.107914)
11. **Zhou, Y.** and McManus, J.F. (2022). Extensive evidence for a Last Interglacial Laurentide Outburst (LILO) event. *Geology*, 50(8), 934-938. doi:[10.1130/G49956.1](https://doi.org/10.1130/G49956.1)
10. **Zhou, Y.**, McManus, J.F., Jacobel, A., Costa, K., Wang, S., and Alvarez Caraveo, B. (2021). Enhanced iceberg discharge in the western North Atlantic during all Heinrich events of the last glaciation. *Earth and Planetary Science Letters*, 564, 116910. doi:[10.1016/j.epsl.2021.116910](https://doi.org/10.1016/j.epsl.2021.116910)
9. Costa, K.M., Hayes, C.T., Anderson, R.F., [and 31 others, including **Zhou, Y.**] (2020). ^{230}Th Normalization: New Insights on an Essential Tool for Quantifying Sedimentary Fluxes in the Modern and Quaternary Ocean. *Paleoceanography and Paleoclimatology*, 35, e2019PA003820. doi:[10.1029/2019PA003820](https://doi.org/10.1029/2019PA003820)
8. Khider, D., Emile-Geay, J., McKay, N.P., [and 89 others, including **Zhou, Y.**] (2019). PaCTS 1.0: A Crowdsourced Reporting Standard for Paleoclimate Data. *Paleoceanography and Paleoclimatology*, 34, 1570-1596. doi:[10.1029/2019PA003632](https://doi.org/10.1029/2019PA003632)
7. Jacobel, A.W., Anderson, R.F., Winckler, G., Costa, K.M., Gottschalk, J., Middleton, J.L., Pavia, F.J., Shoenfelt, E.M., and **Zhou, Y.** (2018).

No evidence for equatorial Pacific dust fertilisation. *Nature Geoscience*, 12, 154-155. doi:10.1038/s41561-019-0304-z

6. Emile-Geay, J., Cobb, K.M., Carre, M., Braconnot, P., Leloup, J., **Zhou, Y.**, Harrison, S.P., Correge, T., McGregor, H.V., Collins, M., Driscoll, R., Elliot, M., Schneider, B., and Tudhope, A. (2016). Links between tropical Pacific seasonal, interannual and orbital variability during the Holocene. *Nature Geoscience*, 9, 168. doi:10.1038/ngeo2608

5. Dee, S., Noone, D., Buening, N., Emile-Geay, J., and **Zhou, Y.** (2015). SPEEDY-IER: A fast atmospheric GCM with water isotope physics. *Journal of Geophysical Research - Atmospheres*, 120, 73–91. doi:10.1002/2014JD022194

4. Caballero-Gill, R.P., Libarkin, J., Meyers, S.R., Hinnov, L., McCallum, C., Lisiecki, L.E., Malinverno, **Zhou, Y.**, Segessenman, D., A., Kochen, I., Hobart, B., Ajibade, R.A., Kinney, S., Olsen, P., and Omar, H. Addressing Barriers in Postdoctoral Recruitment and Application Processes: An Equity-based Framework. *Nature Communications*. In revision.

3. Middleton, J., Pavia, F., Anderson, R.F., Schwartz, R., Fleisher, M., Lao, Y., **Zhou, Y.**, Kinsley, C., Schaefer, J., Frank, M., and Winckler, G. Oceanographic and climatic controls of meteoric ¹⁰Be fluxes to seafloor sediments: A global synthesis. *Quaternary Science Reviews*. In revision.

2. **Zhou, Y.**, Lakhani, K.Q., Lynch-Stieglitz, J., Gebbie, G., and Pallone, C.T. Deepening and double-ridge structure of the Last Glacial Maximum tropical Pacific thermocline. In prep.

1. Lisiecki, L.E., Gleason, L., Hobart, B., and **Zhou, Y.** Obliquity and Precession Influences on North Atlantic Ice Rafting Events of the Late Pleistocene. In prep.

NON-PEER- REVIEWED PUBLICATIONS

2. **Zhou, Y.** and McManus, J.F. (2024). Is collapse of the Atlantic Ocean circulation really imminent? Icebergs' history reveals some clues. *The Conversation*. <http://bit.ly/4e9uUA4>

1. Lisiecki, L.E., **Zhou, Y.**, Kochhann, K.G.D., Holbourn, A., Arrigoni, A., and members of the TIMES Working Group 3. Chemostratigraphy using Oxygen Isotopes in Foraminiferal Calcite. *Past Global Changes Magazine*. In review.

AWARDS & FUNDING

NSF OCE-2508421 Collaborative Research: Reconstructing Ocean Sedimentation Rate Variability to Improve Age Estimates for Paleoclimate Data. (\$308,843) 2025

Note: Authored a major section of the proposal. Ineligible to be a PI due to UCSB rules.

Editor's Citation for Excellence in Refereeing, Geophysical Research Letters	2024
NOAA Climate and Global Change Fellowship (alternate)	2022
Cushman Foundation Johanna M. Resig Fellowship (\$30,000)	2021
IODP Schlanger Fellowship (\$30,000)	2020
Columbia University GSAS Dean's Fellowship	2016
Woods Hole Oceanographic Institution Summer Student Fellowship	2015

- CONFERENCES (FIRST AUTHOR & CONVENOR)
18. **Zhou, Y.**, McManus, J.F., Pallone, C., Kenna, T.C., Weinstein, G.A., and Garcia, H. Abrupt weakening of deep Atlantic circulation at the last glacial inception. *2025 AGU Fall Meeting* (Invited speaker)
 17. Omar, H., Hinnov, L.A., Villa, A., **Zhou, Y.**. Cyclostratigraphy and Astronomical Forcing of Earth's Paleoclimate System. *2025 AGU Fall Meeting* (Convenor)
 16. **Zhou, Y.** and McManus, J.F. Heinrich event ice discharge and the fate of the Atlantic Meridional Overturning Circulation. *2024 AGU Fall Meeting* (Invited speaker)
 15. Villa, A., Hinnov, L.A., Li, M., Omar, H., Wu, H., **Zhou, Y.**, Wu, Y. Astronomical Forcing of Earth's Paleoclimate System. *2024 AGU Fall Meeting* (Convenor)
 14. **Zhou, Y.** and McManus, J.F. Last interglacial Laurentide outburst (LILO) event and ice volume distributions during the last interglacial. *2024 PALSEA-Next*
 13. **Zhou, Y.**, Lisiecki, L.E., Meyers, S. A new probabilistic, orbitally tuned Pleistocene stack of benthic $\delta^{18}\text{O}$. *2023 AGU Fall Meeting* (Invited speaker)
 12. **Zhou, Y.**, Lisiecki, L.E. Advances in Our Understanding of Climate Change During the Plio-Pleistocene 41-kyr World. *2023 AGU Fall Meeting* (Convenor)
 11. **Zhou, Y.**, Lisiecki, L.E., Rand, D., Hobart, B., Lee, T., Gebbie, G., and Lawrence, C.E. Revisiting the early Pleistocene "41-kyr world" benthic $\delta^{18}\text{O}$ stack - an Atlantic-Pacific divergence during 1.8-1.9 ma. *2023 AGU Fall Meeting*

10. **Zhou, Y.** and McManus, J.F. Heinrich event ice discharge and the fate of the Atlantic Meridional Overturning Circulation. *2023 Comer Climate Conference*
9. **Zhou, Y.**, Lisiecki, L.E., Rand, D., Hobart, B., Lee, T., Gebbie, G., and Lawrence, C.E. Revisiting Pleistocene benthic $\delta^{18}\text{O}$ stacks with BIGMACS. *2022 AGU Fall Meeting*
8. **Zhou, Y.** and McManus, J.F. Authigenic uranium deposition in the glacial North Atlantic - implications for oxygenation, carbon storage, and deep water mass geometry. *2022 Comer Climate Conference*
7. **Zhou, Y.** and McManus, J.F. Extensive evidence for a last interglacial Laurentide outburst (LILO) event. *2022 PAGES OSM*
6. **Zhou, Y.** and McManus, J.F. Glacial carbon storage and water mass geometry in the North Atlantic. *2021 AGU Fall Meeting*
5. **Zhou, Y.** and McManus, J.F. Extensive evidence for the Last Interglacial analog of the 8.2 ka event. *2021 Comer Climate Conference*
4. **Zhou, Y.** and McManus, J.F. A new method of estimating freshwater fluxes during millennial events of the last glaciation. *2020 Comer Climate Conference*
3. **Zhou, Y.** and McManus, J.F. Heinrich Events 3 and 6 as Events of Increased Ice-Rafted Deposition. *2019 Goldschmidt*
2. **Zhou, Y.**, Oppo, D., Gebbie, G., and Thornalley, D. Magnitude of the Suess Effect in North Atlantic - a Study of Foraminifera and Transient Tracer Simulations. *2016 AGU Ocean Sciences Meeting*
1. **Zhou, Y.**, Paterson, S., Pablo, A.H., Cao, W. and Ratschbacher, B. An Isostatic Mass Balance Model of Continental Arcs and Its Application to Paleozoic-Mesozoic Argentinean Cordilleran Orogenic Systems. *2016 GSA Cordilleran Section meeting*

INVITED SEMINARS	Climate+Paleo Seminar, Woods Hole Oceanographic Institution	2024
	Department of Oceanography Seminar, University of Alaska Fairbanks	2024
	Departmental Seminar, Grinnell College	2023
	Paleo/Environmental Seminar, University of Southern California	2023
	Earth Research Institute Climate Seminar, UC Santa Barbara	2023
	Whole Earth Seminar, UC Santa Cruz	2023
	Paleoclimate working group meeting, NCAR	2022

	International Quaternary Webinar, UMass Amherst	2022
	Climate-geochemistry departmental seminar, Max Planck Institute for Chemistry	2021
SEAGOING EXPERIENCE	<i>R/V Roger Revelle</i> , UNOLS Coring PI training	Aug. 22 – Sept. 1, 2022
	<i>R/V Tioga</i> , plankton towing and grab coring training	Jul. 17, 2015
PROFESSIONAL SERVICES	Member, AGU Paleo & Paleo Early Career Working Group	2024 –
	Chair, LDEO graduate student committee	2019 – 2020
	Volunteer, LDEO Wally Broecker Symposium	2019
	Orientation co-chair, LDEO graduate student committee	2018 – 2019
	Member, LDEO committee on professional conduct	2018 – 2022
	Co-organizer, LDEO geochemistry seminar committee	2017 – 2019
	Co-organizer, LDEO first year colloquium	2017
	President, USC Sigma Gamma Epsilon	2015 – 2016
	Secretary, —	2014
	Reviewer for <i>NSF (ad hoc & panel)</i> , <i>Nature Geoscience</i> , <i>Science Advances</i> , <i>Nature Communications</i> , <i>Environmental Research Letters</i> , <i>Earth and Planetary Science Letters</i> , <i>Geophysical Research Letters</i> , <i>Paleoceanography and Paleoclimatology</i> , <i>Climate of the Past</i> , <i>Quaternary Science Reviews</i> , <i>Geo-Marine Letters</i> , <i>Marine Geology</i> , <i>Scientific Reports</i> , and <i>Quaternary Research</i>	
TEACHING AND OUTREACH	Author, “ Climate Bracelet ”, UCSB Family Ultimate Science Exploration	2025
	Facilitator, UCSB Family Ultimate Science Exploration	2022 – 2025
	Domain Expert, ClimateMatch Academy	2023
	Guest lecturer, UCSB Earth 4 Introduction to Oceanography	2023
	Teacher, Columbia University Girls Who Code	2018 – 2022
	Guest lecturer, Columbia University EESCW4920 Paleooceanography	2022
	Teacher, Columbia University COVID Volunteer Tutor Corps	2020
	Foundational Track certification, Teaching Development Program, the Center for Teaching and Learning at Columbia University	2020
	Guest lecturer, Columbia University EESCW2100 Earth’s Environmental Systems: Climate System	2018

Teaching Assistant, Columbia University EESCW2100 Earth's
Environmental Systems: Climate System 2017 – 2018

Volunteer, Columbia University Girls Science Day 2018 – 2019

Volunteer, Lamont Open House 2016 – 2019

Oceanography Teacher, Lenicia B. Weemes Elementary School, Joint
Education Program 2012

Mentor of Alyson Churchill (Colby College, 2018; now PhD student at
Oregon State University), Annemarie Pillsbury (Dutchess Community
College, 2018; now graduate student at University at Buffalo), Miah
Cohall (Manhattan College, 2019; now Assistant Engineer at Hazen and
Sawyer), Cassandra Bartels (Barnard College, 2019; now PhD student at
the WHOI-MIT join program), Ellen May Jorgensen (Syracuse University,
2021; now PhD student at Brown University), Herman Garcia (The City
College of New York, 2021; now PhD student at University of Colorado,
Boulder), Ariana Paul (Barnard College, 2021), Chandler Morris
(Columbia University, 2021; now PhD student at Brown University),
Diego Sevilla (UCSB, 2023-2024; senior thesis primary advisor), Andrew
Toteda (UCSB, 2024-2025; senior thesis primary advisor)

PROGRAMMING Python (expert), Matlab (expert), C++ (proficient), Java (proficient),
LANGUAGES Julia (competent), Fortran (competent), R (competent)